

Spatially-Enriched, Retrieval-Augmented and Expert-Specialized Missing-Modality Generation for Multimodal Sentiment Analysis

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I. PROJECT DETAILS

Criterion. This project follows the criterion of extending an existing paper with an available codebase by addressing identified limitations and proposing measurable improvements. We will build upon *Multimodal Prompt Learning with Missing Modalities for Sentiment Analysis and Emotion Recognition* (MPLMM) [1].

The project will proceed incrementally:

- 1) Reproduce the MPLMM baseline.
- 2) Introduce 2D spatial feature convolutions to preserve spatial facial information before visual representations are supplied to the multimodal architecture.
- 3) Extend missing-modality reconstruction with Retrieval-Augmented Generation (RAG) by providing relevant complete training examples as additional reconstruction context.
- 4) Explore Residual RAG to determine whether reconstruction errors exhibit useful locality and can be retrieved for reconstruction correction.
- 5) Introduce a Direct Mixture-of-Experts Missing-Modality Generator (MoE-MMGM) in which specialized experts directly generate the missing modality using available modalities, retrieved context, and generative prompts.

The overall progression is

MPLMM $\rightarrow$ 2D Spatial $\rightarrow$ RAG
 $\rightarrow$ Residual RAG $\rightarrow$ MoE-MMGM.

The three primary proposed components are therefore 2D spatial feature convolution, retrieval-augmented reconstruction, and direct MoE-based missing-modality generation. Residual RAG is treated as an exploratory extension of the retrieval component.

Team Composition and Individual Contributions.

- **Lanka Devi Satwika (B24DS013):** MPLMM reproduction, dataset preparation, baseline evaluation, and 2D spatial CNN.
- **Siddhi Singh (B24DS508):** 2D spatial feature convolution, visual preprocessing pipeline, spatial-feature evaluation, Retrieval system (RAG), and Residual RAG exploration.

- **Vidit Shrimali (B24DS038):** Retrieval system, Residual RAG exploration, Direct MoE-MMGM integration, ablation studies, and overall evaluation.

II. PROBLEM STATEMENT

Multimodal sentiment analysis combines text, audio, and visual modalities, but real-world inputs are often incomplete because one or more modalities may be unavailable. MPLMM addresses this problem by reconstructing the missing modality from the available modalities using its Missing Modality Generation Module (MMGM) and then providing the generated representation to the multimodal Transformer [2].

However, the current implementation leaves three important opportunities for improvement.

A. Limited Spatial Information

The existing MPLMM pipeline operates on pre-extracted visual representations. This makes the visual pipeline convenient, but spatial facial structure is not explicitly modeled by the MMGM pipeline. Fine-grained cues such as localized facial movements may therefore be underrepresented. We propose introducing a 2D CNN before the visual representation becomes the feature-vector sequence, allowing spatial information to be encoded before multimodal reconstruction.

B. Limited Reconstruction Context

MPLMM primarily reconstructs the missing modality from the modalities available in the current sample. When those modalities provide insufficient evidence, the generator has no direct access to analogous complete examples from the training distribution. We propose RAG-based context ingestion, retrieving relevant complete samples and providing their representations as additional evidence for reconstruction. We will further explore Residual RAG, where the retrieved information represents reconstruction-error patterns rather than only modality representations.

C. A Single Generation Function

MPLMM's MMGM uses a common generation mechanism for different samples and missing-modality patterns. However, reconstruction requirements can vary substantially

across samples. We therefore propose a Direct MoE-MMGM, where multiple specialized experts generate candidate missing-modality representations and a router dynamically combines them.

Thus, our proposed progression is

Spatial Information $\rightarrow$ Retrieved Context

$\rightarrow$ Specialized Generation.

Research Problem. *Can enriching the visual representation with spatial information, supplying instance-level evidence through retrieval, and using specialized generation experts improve missing-modality reconstruction and downstream multimodal sentiment prediction?*

III. METHODOLOGY

A. MPLMM Baseline

MPLMM contains a Missing Modality Generation Module (MMGM) that reconstructs a missing modality using the modalities available in the current sample and a learned generative prompt. The reconstructed representation is subsequently supplied to the multimodal Transformer together with missing-signal information. For example, when Audio is missing,

$$\hat{x}_A = G(x_V, x_T, P_G^A) \quad (1)$$

where x_V and x_T denote the Video and Text representations, P_G^A is the learned generative prompt for Audio, G is the MMGM, and $\hat{x}_A$ is the reconstructed Audio representation. We first reproduce MPLMM as the controlled baseline against which each proposed component will be evaluated. The baseline architecture is shown in Fig. 1.

B. 2D Spatial Feature Convolution

We modify the visual preprocessing pipeline by applying a lightweight 2D CNN directly to the spatial visual input before the visual data is converted into the feature-vector sequence used by MPLMM. For a video frame at time t ,

$$X_V^t \in \mathbb{R}^{C \times H \times W} \quad (2)$$

where X_V^t is the raw visual frame, C is the number of input channels, and H, W are the spatial height and width.

The spatial feature map is obtained as

$$F_V^t = \sigma(W_{2D} * X_V^t + b) \quad (3)$$

where W_{2D} is the learnable 2D convolution kernel, b is the convolution bias, $*$ denotes 2D convolution, and σ is the activation function.

The resulting feature map is converted into the representation required by MPLMM:

$$x_V^t = \text{Proj}(\text{Flatten}(F_V^t)) \quad (4)$$

where F_V^t is the extracted spatial feature map, $\text{Flatten}(\cdot)$ converts it into a vector, $\text{Proj}(\cdot)$ is a learnable projection, and x_V^t is the resulting visual representation at time t . Thus

Raw Frame $\rightarrow$ 2D CNN $\rightarrow$ Spatial Features
 $\rightarrow$ Projection $\rightarrow$ MPLMM.

The objective is to preserve spatial facial geometry and localized visual cues before multimodal reconstruction. The modified visual pipeline is shown in Fig. 2.

What we test: whether spatially enriched visual representations improve missing-modality reconstruction and downstream sentiment prediction.

C. Retrieval-Augmented Reconstruction

We extend the MMGM by providing retrieved complete training examples as additional reconstruction context. Instead of

$$\hat{x}_m = G(x_{\text{avail}}, P_G^m) \quad (5)$$

we use

$$\hat{x}_m = G(x_{\text{avail}}, R_m, P_G^m) \quad (6)$$

where x_{avail} denotes the representations of the available modalities, R_m denotes the retrieved representations associated with the missing modality m , P_G^m is the generative prompt for modality m , G is the missing-modality generator, and $\hat{x}_m$ is the reconstructed representation.

The retrieval query is constructed from the available modalities,

$$q = f(x_{\text{avail}}) \quad (7)$$

where q is the retrieval query and $f(\cdot)$ maps the available modalities into the retrieval space. The top- K complete training samples are then retrieved:

$$R_q = \text{TopK}(q, \mathcal{M}) \quad (8)$$

where $\mathcal{M}$ is the memory bank of complete training samples, K is the number of retrieved samples, and R_q is the retrieved set. We will initially evaluate $K \in \{1, 3, 5, 10\}$.

The objective is to determine whether instance-level evidence from complete samples can improve reconstruction when the available modalities alone are insufficient. The architecture is shown in Fig. 3.

D. Residual RAG Exploration

We additionally investigate whether retrieval can provide information about the systematic reconstruction errors of the base generator. For a complete training sample i ,

$$r_i = x_{m,i} - \hat{x}_{m,i}^{(0)} \quad (9)$$

where $x_{m,i}$ is the ground-truth representation of missing modality m , $\hat{x}_{m,i}^{(0)}$ is its reconstruction from the base generator, and r_i is the resulting reconstruction residual.

For a new sample, residuals from retrieved similar samples are aggregated:

$$\Delta x_m = \sum_{i \in R_q} w_i r_i \quad (10)$$

where R_q is the retrieved sample set, r_i is the residual associated with retrieved sample i , and w_i is its normalized similarity weight. A possible corrected reconstruction is then

$$\hat{x}_m = \hat{x}_m^{(0)} + \lambda \Delta x_m \quad (11)$$

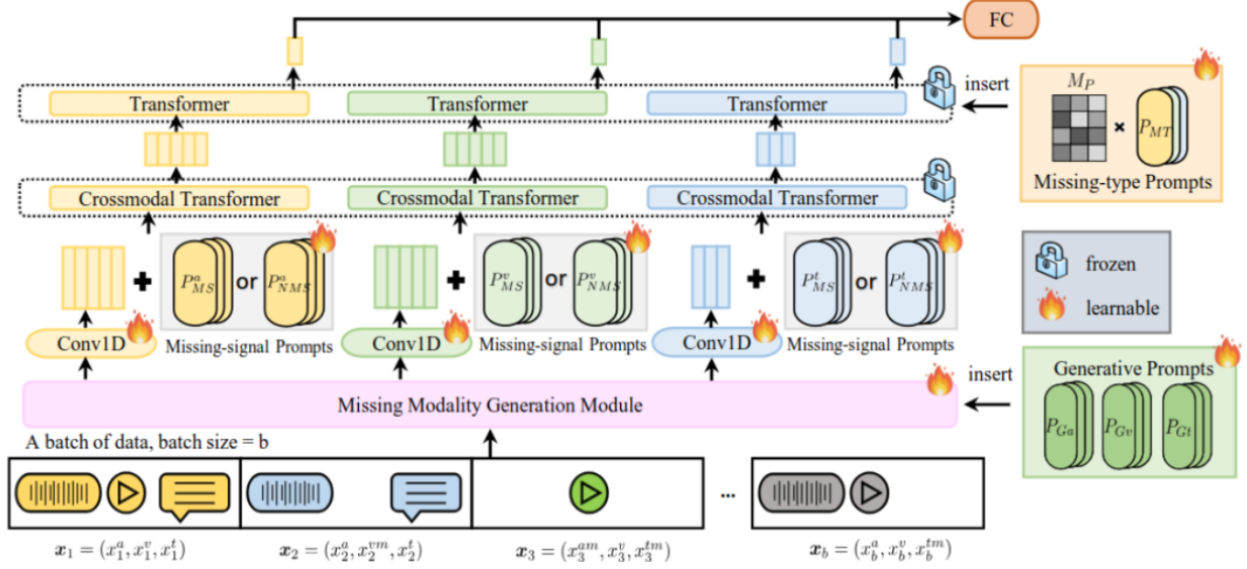


Fig. 1. MPLMM baseline architecture.

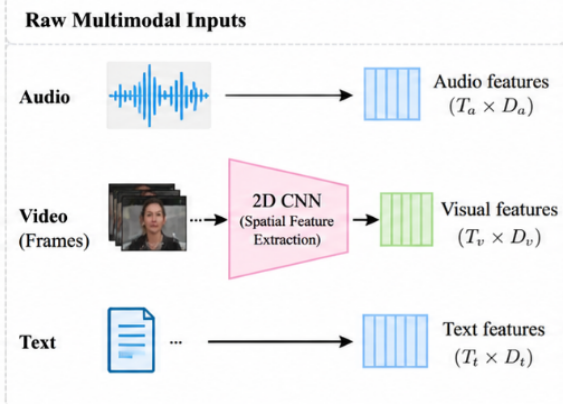


Fig. 2. 2D spatial feature convolution architecture.

where $\hat{x}_m^{(0)}$ is the initial reconstruction, Δx_m is the retrieved residual correction, and λ controls the strength of the correction.

The first experiment will determine whether similar samples have correlated reconstruction errors. Residual RAG will only be retained if this assumption is supported empirically. The architecture is shown in Fig. 4.

E. Direct MoE Missing-Modality Generator

We replace the single reconstruction function with a Direct Mixture-of-Experts Missing-Modality Generator (MoE-MMGM). The generator receives the available modalities, retrieved context, and generative prompt:

$$p = \text{Router}(x_{\text{avail}}, R_m, P_G^m) \quad (12)$$

where $p = [p_1, \dots, p_K]$ contains the routing weights assigned to the K generation experts. Each expert produces a candidate

reconstruction:

$$\hat{x}_m^{(k)} = G_k(x_{\text{avail}}, R_m, P_G^m) \quad (13)$$

where G_k is the k -th specialized generation expert and $\hat{x}_m^{(k)}$ is its candidate reconstruction. The final missing-modality representation is

$$\hat{x}_m = \sum_{k=1}^K p_k \hat{x}_m^{(k)} \quad (14)$$

where p_k is the routing weight for expert k , $\hat{x}_m^{(k)}$ is the output of expert k , and $\hat{x}_m$ is the final reconstructed modality.

The experts are intended to specialize in different missing-modality generation mappings, while the router determines their contribution based on the missing-modality condition and available evidence. The objective is to test whether explicit expert specialization can improve missing-modality generation over a single MMGM. The architecture is shown in Fig. 5.

IV. RESOURCE AND DATASET AVAILABILITY

A. Datasets

The project will use the multimodal datasets used by MPLMM: CMU-MOSEI [4], CMU-MOSI [3], IEMOCAP [5] and CH-SIMS [6]. These datasets contain text, audio, and visual information and allow controlled simulation of missing-modality conditions. The initial MPLMM reproduction will use the same extracted feature representations and experimental setup to maintain a controlled baseline comparison.

For the 2D spatial convolution experiment, the visual preprocessing pipeline will additionally operate on spatial visual inputs before producing the visual representations used by the multimodal model.

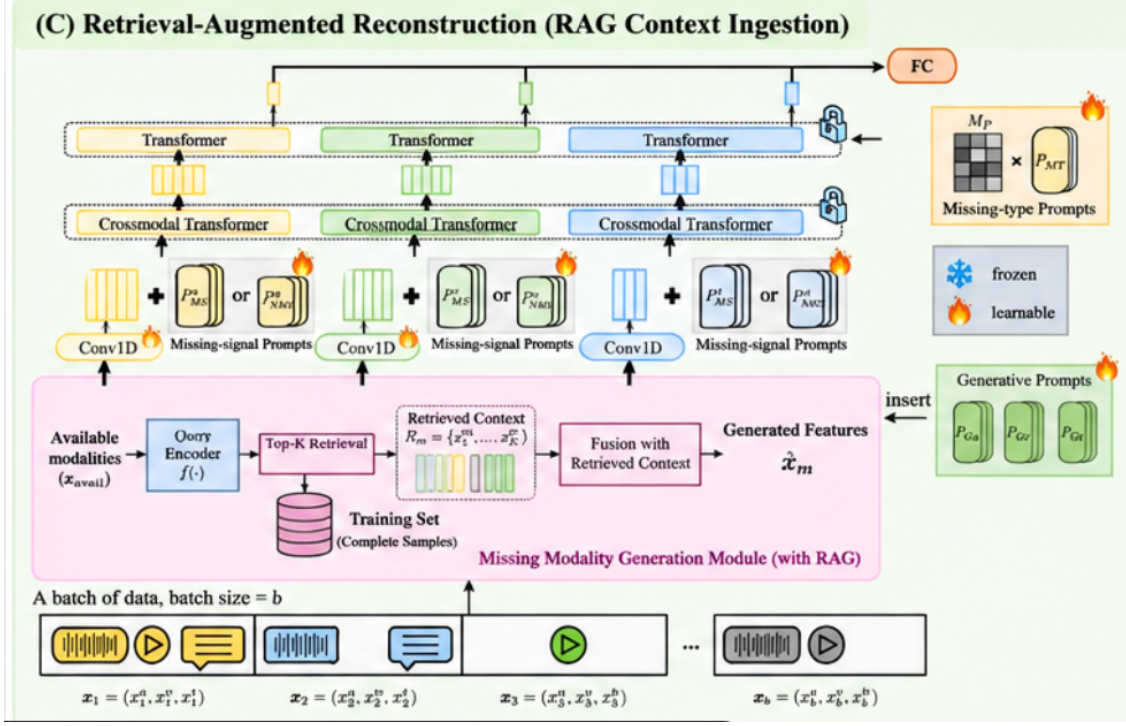


Fig. 3. Retrieval-augmented MMGM.

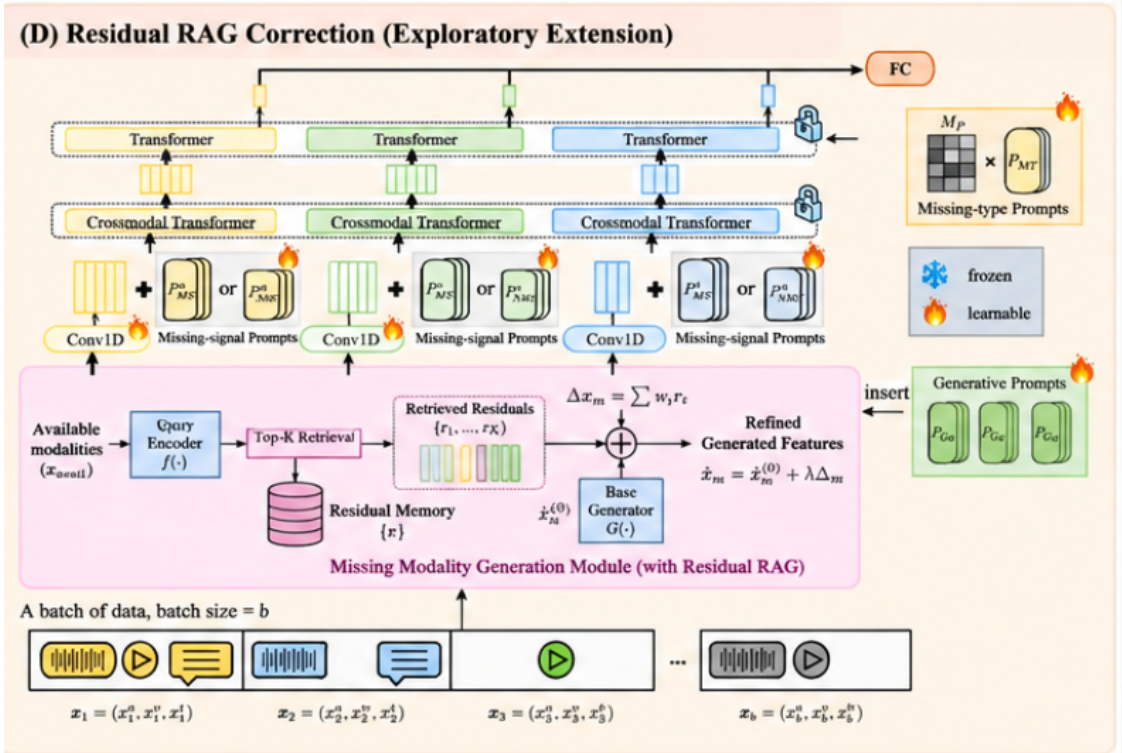


Fig. 4. Residual RAG architecture.

B. Hardware

The project targets an NVIDIA GPU with approximately 8–12 GB VRAM, at least 16 GB system RAM, Python, and

PyTorch. The original MPLMM experiments were conducted on an NVIDIA RTX 3080 with 10 GB VRAM.

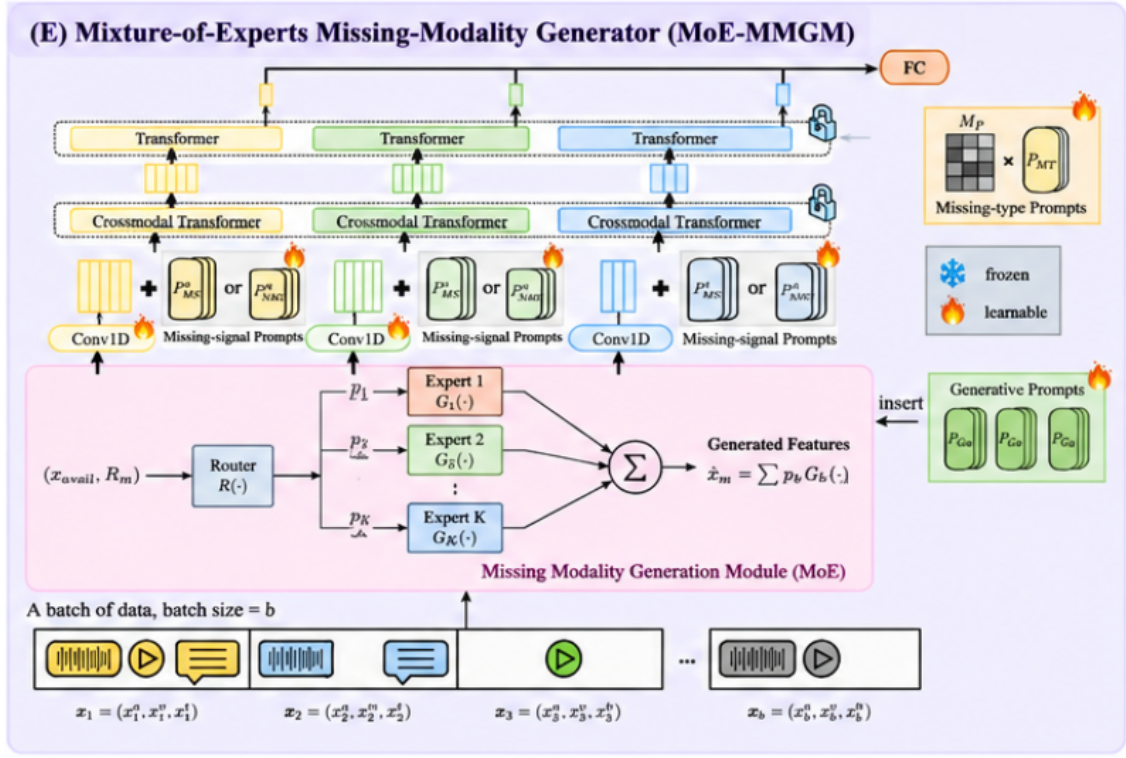


Fig. 5. Direct MoE-MMGM architecture.

C. Additional Resources

- Public MPLMM implementation.
- PyTorch.
- Vector similarity search/index for RAG.
- Complete training-set memory bank.
- Visual frame/spatial feature processing pipeline.
- Reference implementations for MoE and related retrieval methods.

V. LIMITATIONS FOUND

A. Limited Spatial Information in Visual Representations

Limitation: The existing MPLMM pipeline relies on visual feature representations and does not explicitly introduce a dedicated 2D spatial feature-learning stage.

Our approach: We introduce a lightweight 2D CNN before visual feature-vector formation to preserve spatial facial geometry and localized visual information.

Expected improvement: Better visual representations may provide stronger evidence for missing-modality reconstruction and improve downstream sentiment prediction.

B. Limited Instance-Specific Context for Reconstruction

Limitation: MPLMM’s MMGM primarily reconstructs the missing modality from the modalities available in the current sample.

Our approach: We introduce RAG-based reconstruction, retrieving relevant complete training examples and supplying their representations as additional context to the generator.

Expected improvement: Retrieved examples may provide information that is not sufficiently represented in the current incomplete sample, improving reconstruction quality and generalization. We will additionally explore Residual RAG to determine whether reconstruction-error patterns from similar samples can provide useful correction information.

C. Limited Specialization in Missing-Modality Generation

Limitation: The existing MMGM provides a shared generation framework for reconstructing different missing modalities. Different modalities, however, have different representation characteristics and reconstruction mappings.

Our approach: We introduce a Direct MoE-MMGM, where multiple experts specialize in different missing-modality generation functions and a router determines their contribution to the final reconstruction.

Expected improvement: Modality-specialized experts may model reconstruction patterns more effectively than a single shared generation function while maintaining a single forward-pass generation process.

D. Why Our Work Is Unique

The work differs from a standard MPLMM implementation through the combination of three progressively integrated improvements:

- 1) **Spatial enrichment:** 2D convolution preserves visual spatial information before feature-vector formation.

TABLE I
ABLATION DESIGN ISOLATING EACH PROPOSED COMPONENT

Model	2D Spatial	RAG	Resid. RAG	MoE-MMGM
MPLMM	—	—	—	—
+ Spatial	✓	—	—	—
+ Spatial + RAG	✓	✓	—	—
+ Spatial + RAG + Resid.	✓	✓	✓	—
+ Spatial + RAG + MoE	✓	✓	—	✓
Full Model	✓	✓	✓*	✓

*Residual RAG will be included only if its empirical evaluation supports its effectiveness.

- 2) **Context enrichment:** RAG provides relevant complete-instance evidence, with Residual RAG explored as an additional error-correction mechanism.
- 3) **Generation specialization:** Direct MoE-MMGM introduces specialized experts for missing-modality generation.

The work aims to demonstrate improvements in missing-modality reconstruction, downstream sentiment performance, and robustness under varying missing-modality conditions, while keeping the proposed modules lightweight enough for practical experimentation.

VI. EXPECTED OUTCOMES

The principal experiment will isolate the contribution of each proposed component, as listed in Table I.

The combinations shown in the ablation table represent the planned progression; the final combination of later-stage components will be determined empirically based on which components provide consistent improvements.

A. Expected Deliverables

- Reproduced MPLMM baseline.
- 2D spatial visual feature extraction module.
- Retrieval-augmented MMGM.
- Residual RAG experimental module.
- Direct MoE-MMGM.
- Missing-modality robustness evaluation.
- Reconstruction and retrieval analysis.
- Expert-routing analysis.
- Controlled ablation study.
- Trained models/checkpoints.

REFERENCES

- [1] Z. Guo, T. Jin, and Z. Zhao, “Multimodal prompt learning with missing modalities for sentiment analysis and emotion recognition,” in *Proc. ACL*, 2024.
- [2] Y.-H. H. Tsai, S. Bai, P. P. Liang, J. Z. Kolter, L.-P. Morency, and R. Salakhutdinov, “Multimodal transformer for unaligned multimodal language sequences,” in *Proc. ACL*, 2019.
- [3] A. Zadeh, R. Zellers, E. Pincus, and L.-P. Morency, “CMU-MOSI: Multimodal corpus of sentiment intensity and subjectivity analysis in online opinion videos,” 2016.
- [4] A. Zadeh, P. P. Liang, S. Poria, E. Cambria, and L.-P. Morency, “CMU-MOSEI: Multimodal language analysis in the wild,” in *Proc. ACL*, 2018.
- [5] C. Busso *et al.*, “IEMOCAP: Interactive emotional dyadic motion capture database,” *Lang. Resources and Evaluation*, 2008.
- [6] “CH-SIMS: A Chinese multimodal sentiment analysis dataset with fine-grained annotations,” in *Proc. ACL*, 2020.

- [7] RAGPT, “Retrieval-augmented generation for missing modality reconstruction / multimodal prompt learning,” used as inspiration for the retrieval component.

APPENDIX A

TOOLS AND PLATFORMS USED

The proposal was developed using the public MPLMM GitHub implementation as the reference codebase, PyTorch for model development, and a vector similarity search index for the retrieval component. Literature search and idea refinement were supported by general-purpose LLM assistants, and this document was prepared in $\LaTeX$.